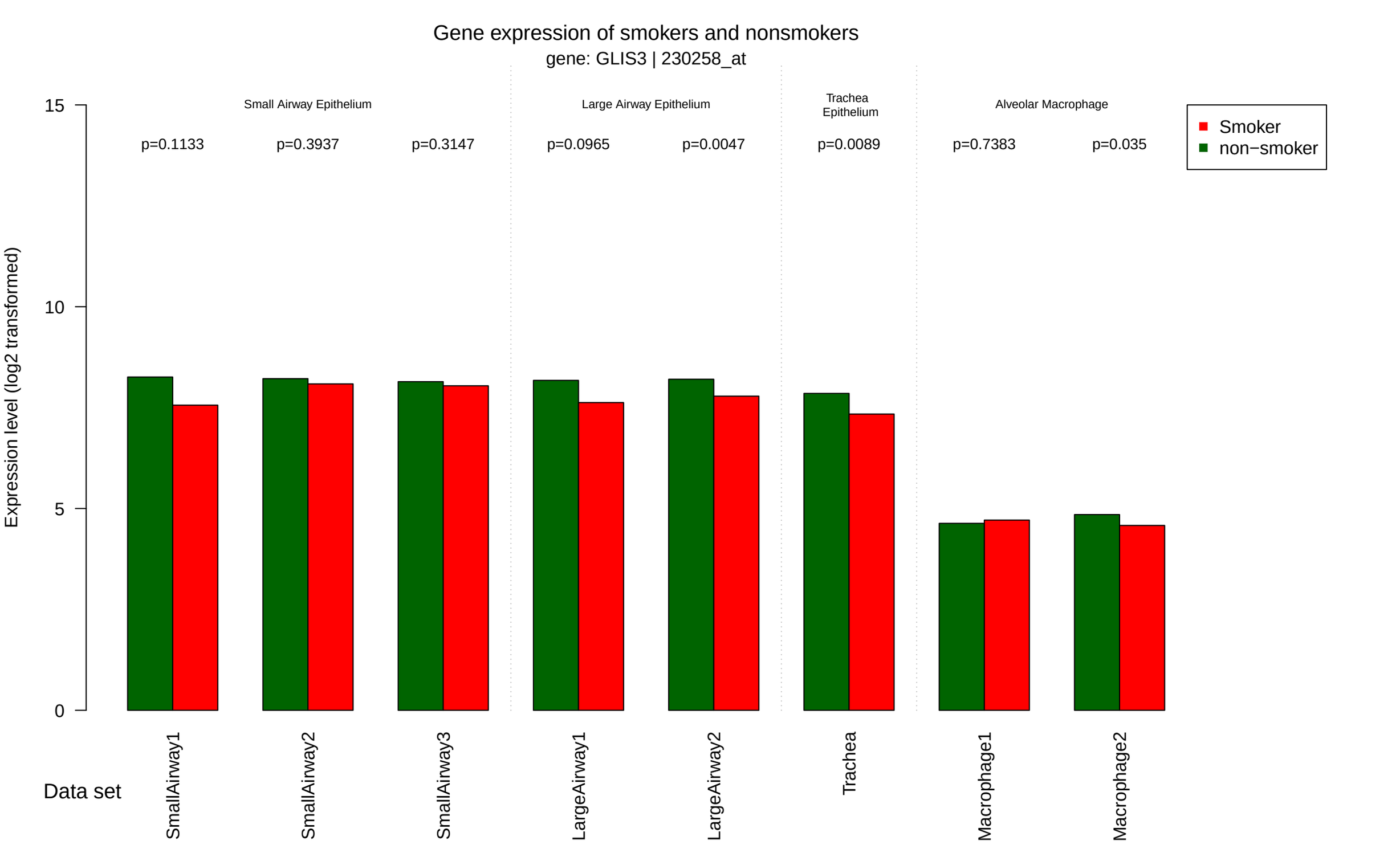
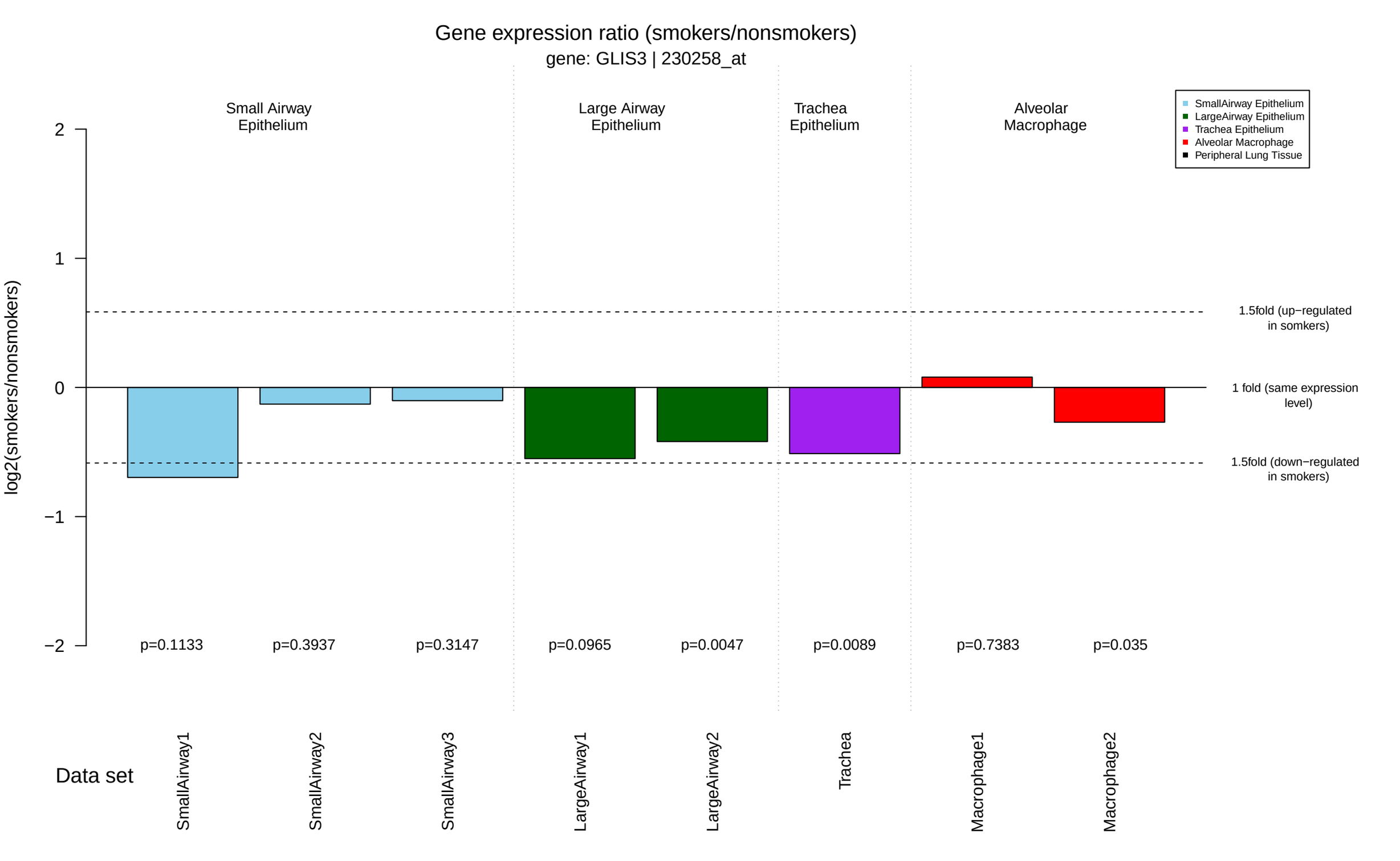
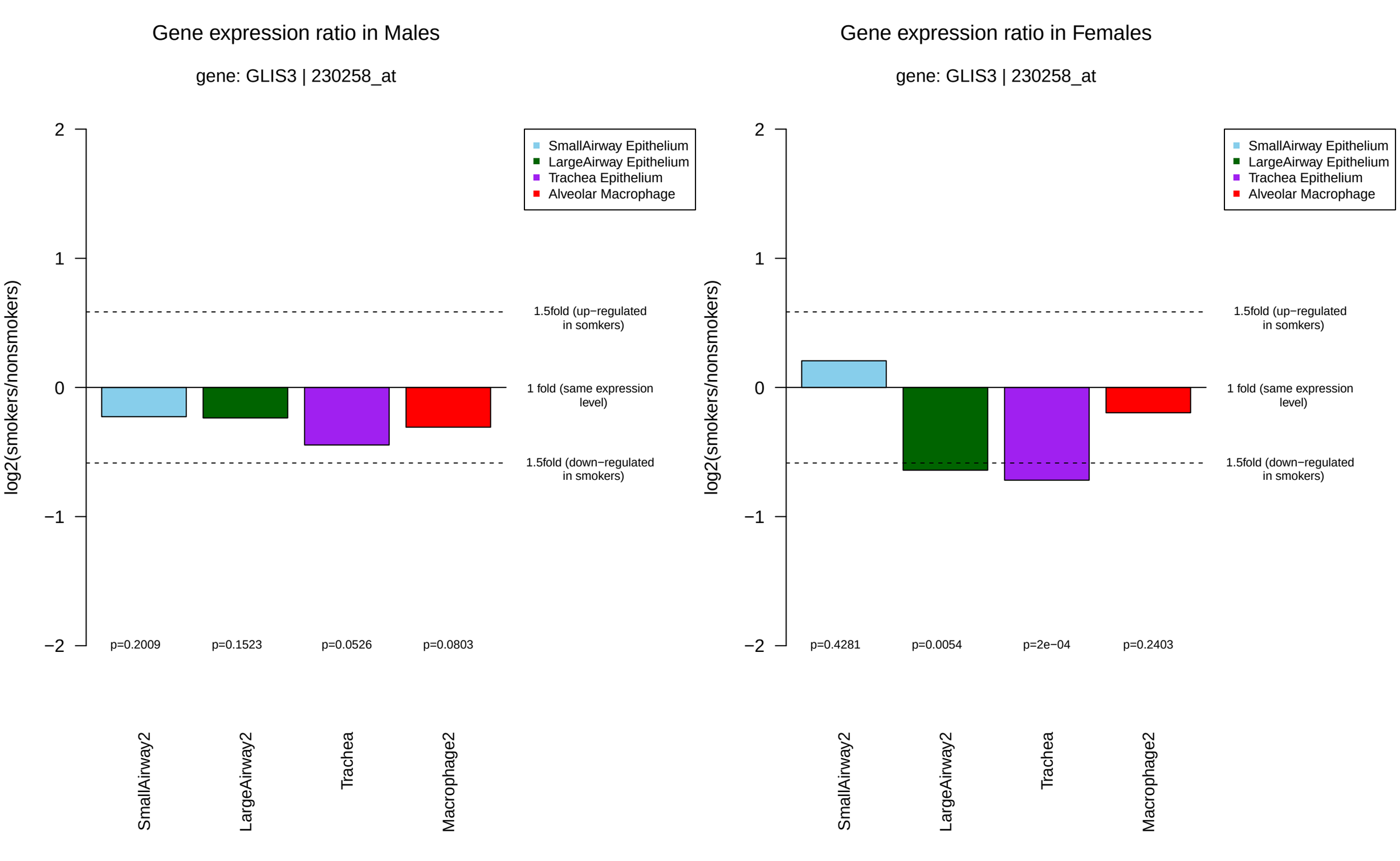
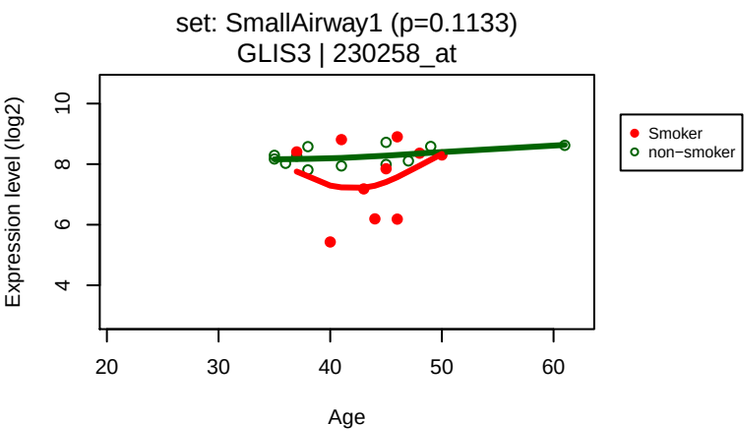
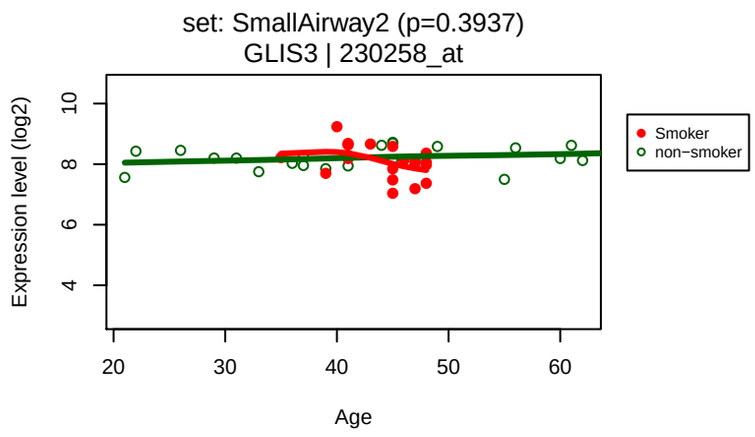
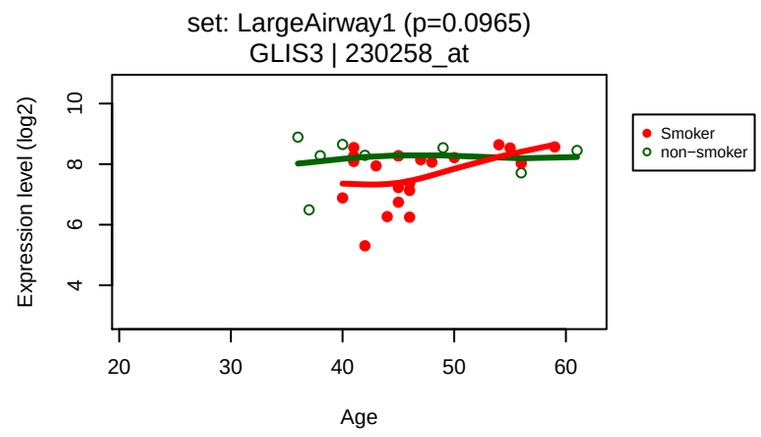

Gene expression of smokers and nonsmokers
gene: GLIS3 | 230258_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1133
p=0.3937
p=0.3147
p=0.0965
p=0.0047
p=0.0089
p=0.7383
p=0.035
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

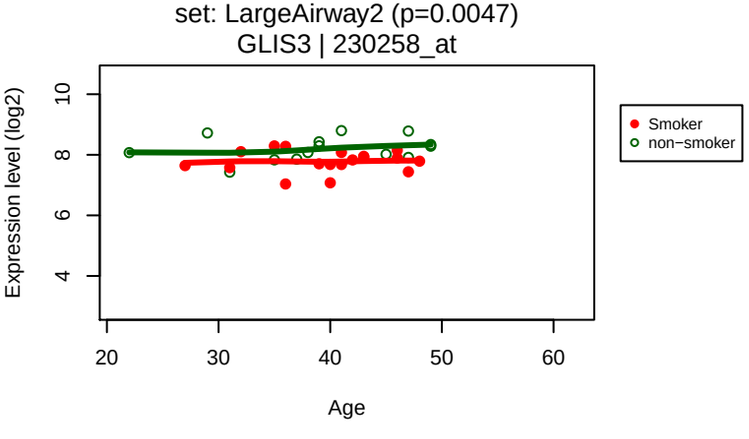

Gene expression ratio (smokers/nonsmokers)
gene: GLIS3 | 230258_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1133
p=0.3937
p=0.3147
p=0.0965
p=0.0047
p=0.0089
p=0.7383
p=0.035
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

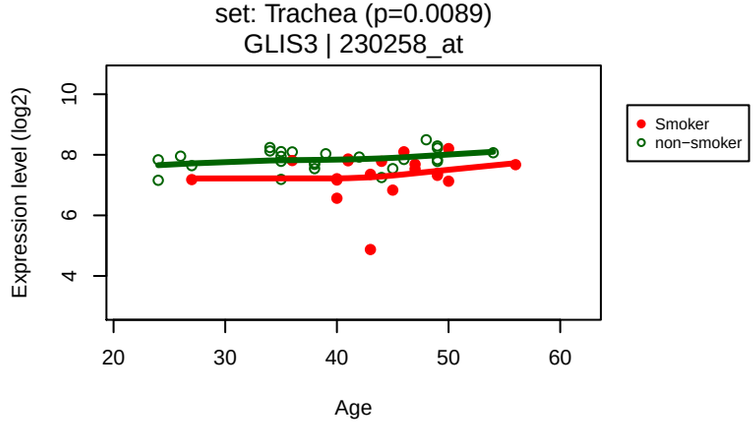

Gene expression ratio in Males
gene: GLIS3 | 230258_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2009
p=0.1523
p=0.0526
p=0.0803
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GLIS3 | 230258_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4281
p=0.0054
p=2e−04
p=0.2403
SmallAirway2
LargeAirway2
Trachea
Macrophage2

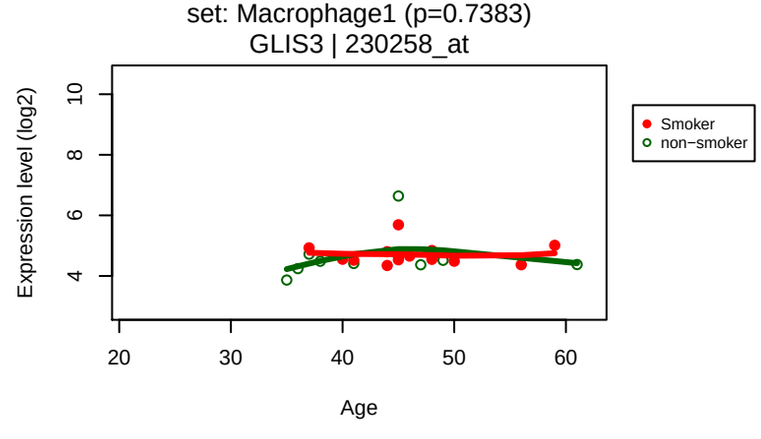

set: SmallAirway1 (p=0.1133)
GLIS3 | 230258_at
Expression level (log2)
Age
Smoker
non−smoker

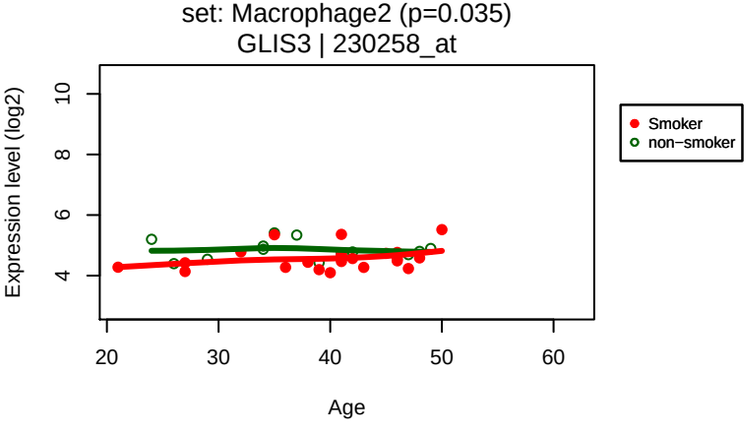

set: SmallAirway2 (p=0.3937)
GLIS3 | 230258_at
Expression level (log2)
Age
Smoker
non−smoker

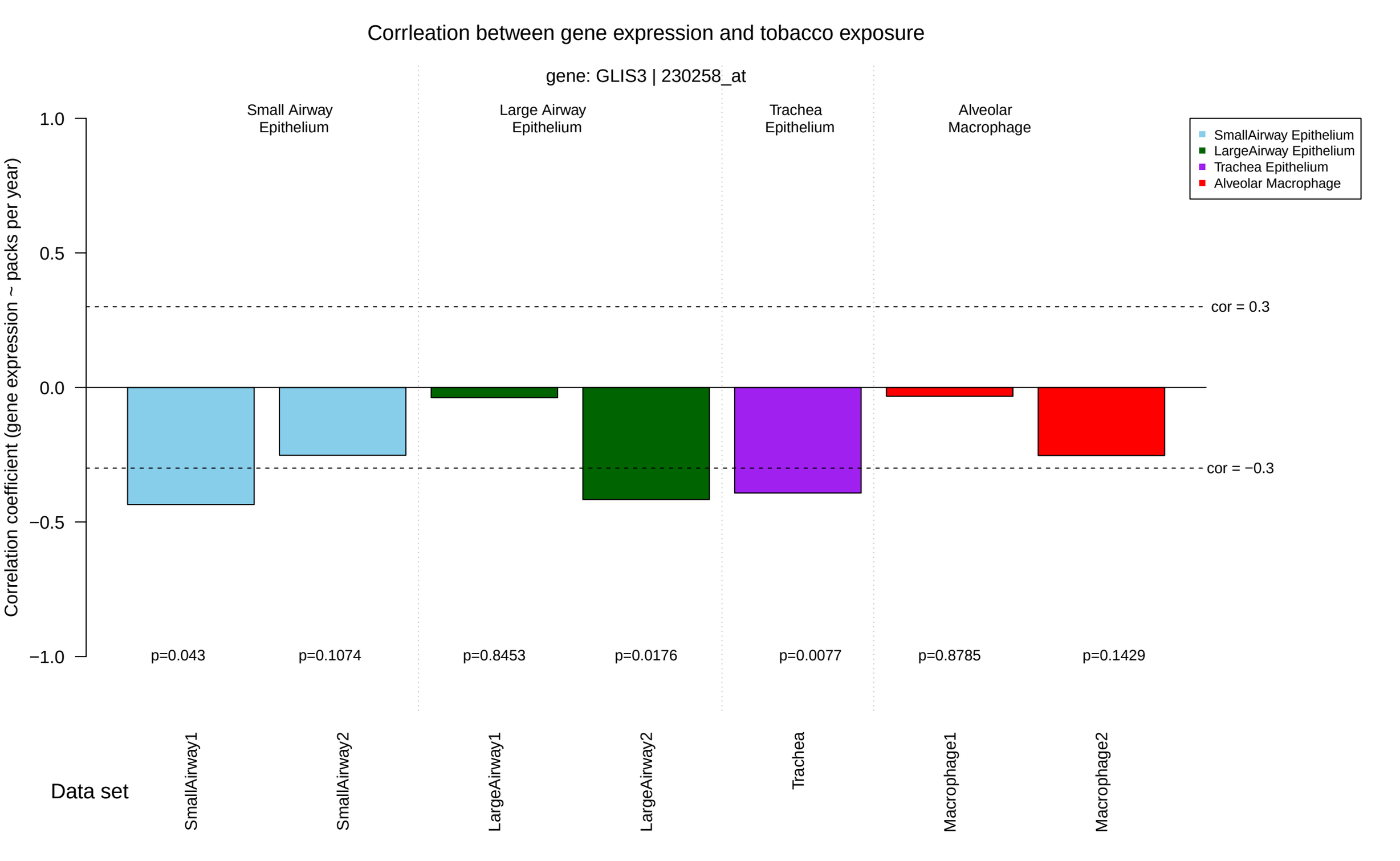

set: LargeAirway1 (p=0.0965)
GLIS3 | 230258_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0047)
GLIS3 | 230258_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0089)
GLIS3 | 230258_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7383)
GLIS3 | 230258_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.035)
GLIS3 | 230258_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GLIS3 | 230258_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.043
p=0.1074
p=0.8453
p=0.0176
p=0.0077
p=0.8785
p=0.1429
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2